(a) One population, twelve paired worlds



10,000 households

Fixed IDs and initial states
Equal household weights

One-off purchase goal

20k | 30k

CNY; defer or purchase

2 goals × 6 paths = 12 worlds

	R0	R1	R2	R3	R4	R5
20k						
30k						

24 months

24

Re-select each month
Independent worlds

2.88M
logical choices
10k × 2 × 6 × 24

Each world starts from the same household states. Executed purchases reduce the remaining goal; the goal never resets each month.

(b) Six external resource paths

Income level as % of original monthly income

		M1–6	M7	M8	M9	M10	M11–24
R0	No shock	100%	100%	100%	100%	100%	100%
R1	Temporary income loss	100%	70%	70%	70%	100%	100%
R2	Persistent income loss	100%	90%	81%	72.9%	72.9%	72.9%
R3	Necessary expense	100%	100%	100%	100%	100%•	100%
R4	Combined shock	100%	70%	70%	70%	100%•	100%
R5	Sustained income gains	100%	110%	121%	133.1%	133.1%	133.1%

• R3 / R4 at M10 only: extra necessary expense = 0.9 × original monthly income.

Same bank product for both goals

Outstanding new principal ceiling: CNY 30,000
5% annual interest / 24 instalments / no fees

(c) Monthly choices and bank feedback

Current shock

Current state

Full household profile
Income and cash
Remaining goal + offer
All executed history
Dated debt contracts

Frozen policy

Monthly choice
YN-mini-MLX
Adjust spending
Wait / cash / finance

Execute ledger

1 Feasible purchase
2 Needs + adjustable
3 Dated payments

Update cash, goal and debt
Carry gaps and arrears

Payments + arrears

Month end: all 10,000 households finish

Bank update in each world

Previous three completed months

$g = \min(1, \max(0.25, 1 - 0.5(1 - c) - 2d))$

c = actual bank payments / payable obligations incl. arrears, capped at 1

d = share of active borrower-months with material 30-day arrears

Next month's offer

Affordable new principal × g

Material 30-day arrears in prior 3 months: offer frozen at 0

g = 0.25: 20k → 5k
Assigned lower bound